INTRODUCTION TO EMBEDDED SYSTEMS

& IoT DEVICE DESIGN EMBEDDED SYSTEMS TASK – 2

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Date: 29 October 2026

Selected mini project: Automatic Light Control Using an LDR Sensor

Working environment: Wokwi Online Simulator

Controller: Arduino UNO

OBJECTIVE

The objective of this project is to interface a Light Dependent Resistor (LDR) with an Arduino UNO and automatically control an LED according to the surrounding light intensity. The project demonstrates analog sensor reading, serial monitoring, and conditional control in an embedded system.

WORKING ENVIRONMENT AND COMPONENTS

The selected environment is the Wokwi online simulator. It allows the circuit and Arduino code to be assembled and tested in a browser without physical components.

COMPONENT	QUANTITY	PURPOSE
Arduino UNO	1	Reads the sensor and controls the LED
LDR / photoresistor	1	Detects changes in surrounding light
10 kΩ resistor	1	Forms a voltage divider with the LDR
LED	1	Shows the output state
220 Ω resistor	1	Limits current through the LED
Connecting wires	As needed	Connect the components

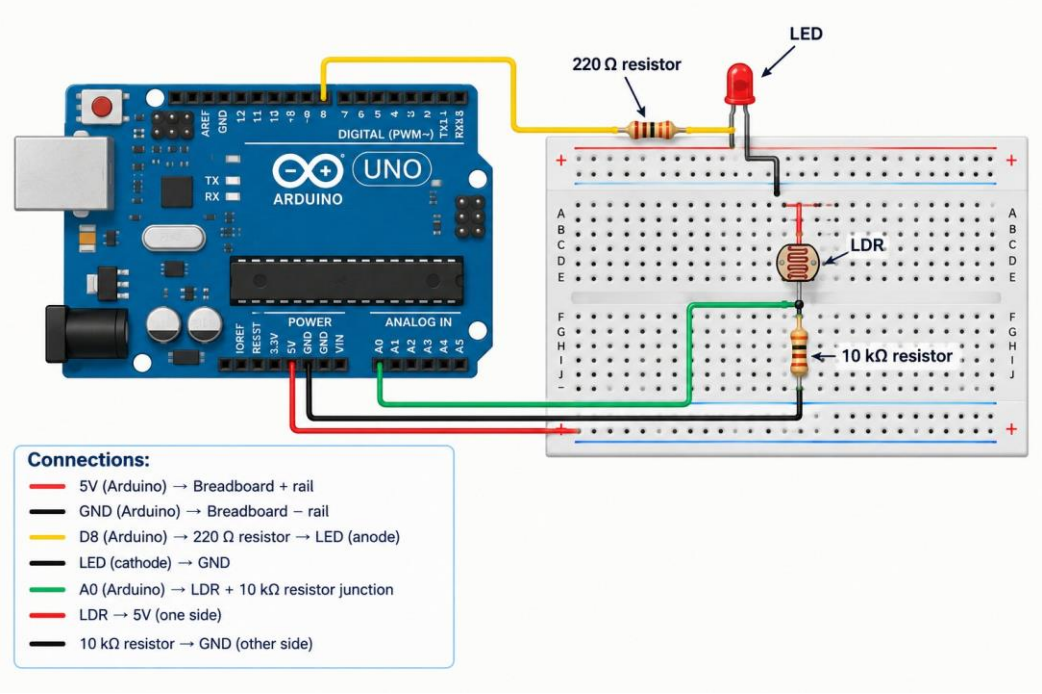
WORKING PRINCIPLE

An LDR is a light-sensitive resistor whose resistance changes with the amount of light falling on it. The LDR and 10 kΩ resistor form a voltage divider. The voltage at the junction changes with the light level, and the Arduino reads it through analog pin A0. The program compares the reading with a threshold value of 500.

Control logic: If the sensor reading is below 500, the LED turns ON. If the reading is 500 or above, the LED turns OFF. This is a starting threshold and may need adjustment after checking the readings in Wokwi.

CIRCUIT DIAGRAM AND CONNECTIONS

COMPONENT PIN	CONNECTION
LDR one side	Arduino 5V
LDR other side / junction	Arduino A0 and one end of the 10 kΩ resistor
Other end of 10 kΩ resistor	Arduino GND
Arduino digital pin 8	220 Ω resistor, then LED anode (long leg)
LED cathode (short leg)	Arduino GND



ARDUINO SOURCE CODE

```
const int ldrPin = A0;
const int ledPin = 8;

int lightValue = 0;
int threshold = 500;

void setup() {
pinMode(ledPin, OUTPUT);
Serial.begin(9600); }

void loop() {   lightValue =
analogRead(ldrPin);

  Serial.print("Light value: ");
Serial.print(lightValue);

if (lightValue < threshold) {
digitalWrite(ledPin, HIGH);
Serial.println(" | Dark: LED ON");   } else
{     digitalWrite(ledPin, LOW);
Serial.println(" | Bright: LED OFF");   }
```

```
delay(500);
```

CODE EXPLANATION

CODE PART	EXPLANATIONS
ldrPin = A0	Selects analog input A0 to read the LDR voltage
ledPin = 8	Selects digital pin 8 to control the LED.
analogRead(ldrPin)	Reads the sensor value, normally between 0 and 1023 on Arduino UNO.
Serial.print / println	Displays the reading and LED status in the Serial Monitor.
if (lightValue < threshold)	Checks whether the reading is below the threshold of 500.
digitalWrite(...)	Turns the LED on (HIGH) or off (LOW).
delay(500)	Waits 500 milliseconds before taking the next reading.

PROCEDURE TO RUN THE PROJECT

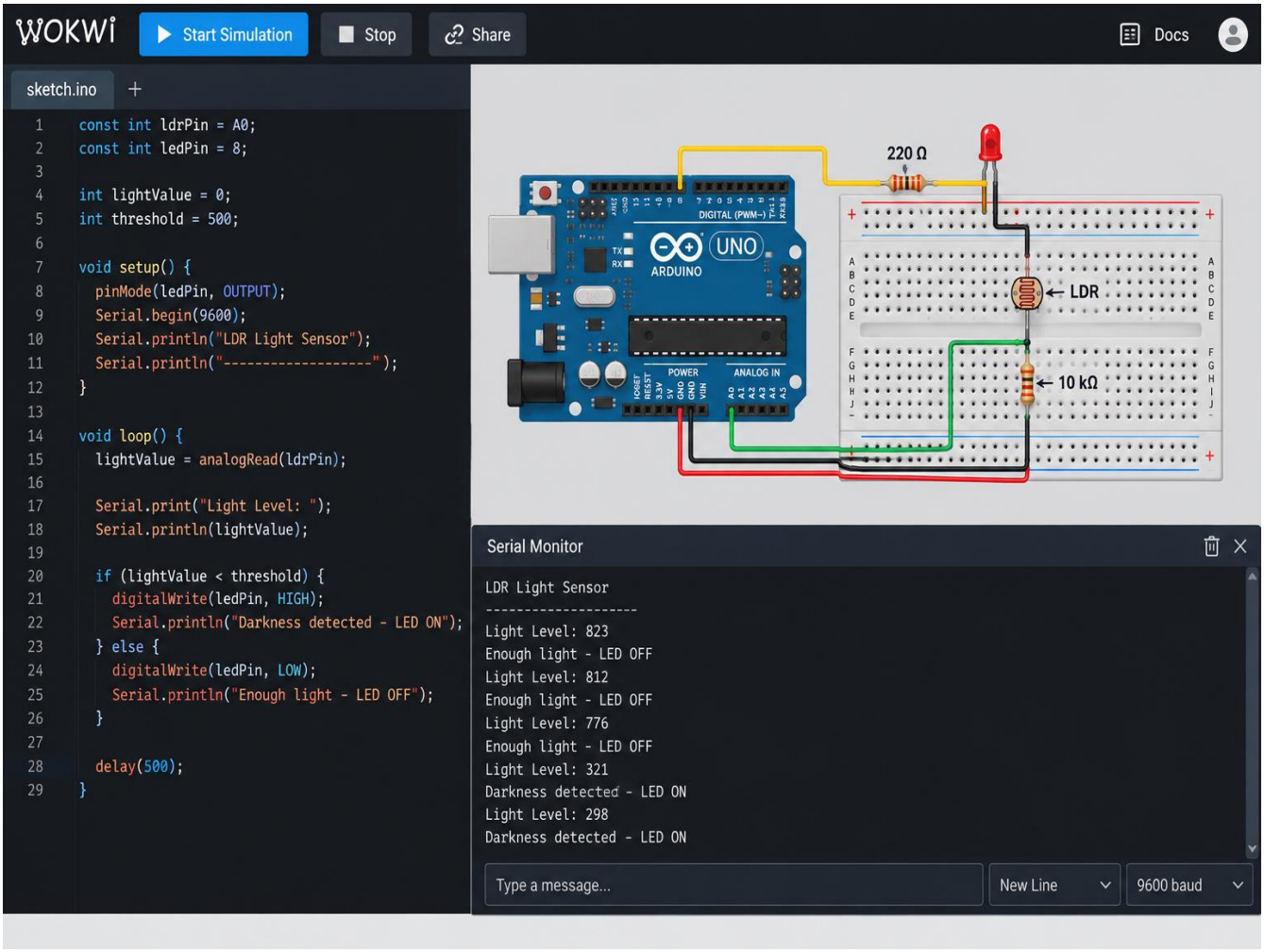
1. Open Wokwi and create a new Arduino UNO project.
2. Add an LDR/photoresistor, a 10 kΩ resistor, an LED, and a 220 Ω resistor.
3. Connect the LDR voltage divider and LED according to Section 4.
4. Replace the sample program in sketch.ino with the source code in Section 5.
5. Start the simulation and open the Serial Monitor.
6. Change the LDR light level in the simulator and observe the sensor reading.
7. Check whether the LED changes state. Adjust the threshold or condition if the readings behave differently.

EXPECTED OUTPUT AND TESTING TABLE

Complete the observed-value column only after running the simulation. The expected LED state below follows the chosen code logic.

TEST	LIGHT CONDITION/ACTION	EXPECTED LED STATE
1	Bright light	OFF if reading ≥ 500
2	Darkness / low light	ON if reading < 500
3	Change light level	State changes when threshold is crossed

CIRCUIT SCREENSHOT



REAL-WORLD APPLICATIONS

- Automatic street lights
- Smart home lighting
- Energy saving lighting systems
- Automatic garden lights

CONCLUSION

This project demonstrates how an LDR can be interfaced with an Arduino UNO to measure changes in light intensity and control an LED automatically. It introduces analog input reading, serial output, and if-else logic, which are basic techniques used in embedded systems and IoT applications.